

Your Cancer Summary Report

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Stage IIA ER+/PR+/HER2- Invasive Ductal Carcinoma with BRCA2
Germline Mutation

Precision Oncology Center — Demonstration Report

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⚠️ DRAFT - FOR CLINICIAN REVIEW

This report has not yet been reviewed by your healthcare team. Please wait for your doctor to discuss these results with you.

At a Glance

Your Diagnosis

Breast Cancer - ER+/PR+/HER2- Invasive Ductal Carcinoma (IDC)

Stage: Stage IIA | **Grade:** Grade 2

You have been diagnosed with Stage IIA breast cancer. This is an early-to-intermediate stage breast cancer that responds to estrogen and progesterone — which is good news, because there are very effective hormone-blocking treatments available. Genomic testing has identified two additional drug targets (PIK3CA and BRCA2) that open further treatment options.

✓ Positive Factors

- ER+/PR+ status means hormone-blocking therapy (tamoxifen) is active and effective
- BRCA2 germline mutation qualifies you for FDA-approved PARP inhibitor therapy (olaparib)
- PIK3CA H1047R mutation is targetable with alpelisib plus fulvestrant (FDA-approved, SOLAR-1)
- CCND1 amplification and CDKN2A deletion both support CDK4/6 inhibitor benefit
- ESR1 strongly clustered in tumor core (Moran's I 0.42) — consistent hormone-driven growth
- All 5 key breast cancer genes significant across RNA, protein, and phospho layers (FDR $q < 7 \times 10^{-11}$)

⚠️ Challenges to Consider

- BRCA2 germline mutation increases lifetime ovarian and other cancer risk — family counseling essential
- MYC amplification (log2 1.45) may contribute to treatment resistance over time
- CDKN2A deletion signals loss of a key cell-cycle checkpoint
- HRD score of 35 is below the 42 clinical threshold — PARP eligibility rests on BRCA2 mutation status, not HRD score

What Your Tests Found

Genetic testing of your tumor revealed the following important findings:

BRCA2

Pathogenic

c.5946delT frameshift (germline, AF=0.50, COSV57492880)

You were born with a change in the BRCA2 DNA repair gene. This raised your cancer risk, but it also makes your tumor sensitive to PARP inhibitor drugs that block a backup repair pathway — essentially forcing cancer cells to self-destruct.

Treatment Option: Olaparib / Niraparib — FDA-approved for BRCA1/2-mutated HER2- breast cancer

PIK3CA

Pathogenic

H1047R missense (somatic, AF=0.42, COSM775)

A common activating change in PIK3CA is turbo-charging a growth signal in your cancer cells. This is one of the best-studied targets in ER+ breast cancer and has an FDA-approved drug combination targeting it directly.

Treatment Option: Alpelisib + fulvestrant — FDA-approved (SOLAR-1 trial) for PIK3CA-mutated HR+/HER2- breast cancer

GATA3

Likely Pathogenic

Frameshift (somatic, AF=0.38, COSV104733902)

A change in the GATA3 gene was found in your tumor. GATA3 defines breast cell identity and is frequently altered in ER+ breast cancer. Its disruption supports the hormone-driven character of your cancer.

Treatment Option: No direct targeted therapy; reinforces endocrine therapy strategy

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MYC

Pathogenic — Copy Number Gain

Amplification, chr8q24 ($\log_2=1.45$, ~5 copies)

Extra copies of the MYC growth driver were found in your tumor. MYC amplification is associated with more aggressive behavior.

Investigational BET inhibitor drugs are being studied to counter this.

Treatment Option: BET inhibitors (investigational) — consider clinical trial enrollment

CCND1

Pathogenic — Copy Number Gain

Amplification, chr11q13 ($\log_2=1.62$, ~6 copies)

Extra copies of the CCND1 cell-cycle accelerator gene were found. CDK4/6 inhibitors specifically target this pathway and are a standard addition to hormone therapy in ER+ breast cancer.

Treatment Option: Palbociclib / Ribociclib / Abemaciclib — FDA-approved for ER+/HER2- breast cancer

CDKN2A

Pathogenic — Copy Number Loss

Deletion, chr9p21.3 ($\log_2=-1.65$, ~1 copy)

A copy of the CDKN2A braking gene (p16) is missing from your tumor cells. Without this brake, the cell-division accelerator runs unchecked — which is exactly what CDK4/6 inhibitors are designed to stop.

Treatment Option: Supports CDK4/6 inhibitor use; CDKN2A loss may predict enhanced response

Your Tumor Environment

Advanced spatial analysis shows how different cells are organized in your tumor:

Tumor Microenvironment

Visium spatial transcriptomics across 900 spots in 7 tissue regions confirms dominant ER-pathway activity in the tumor core. ESR1 mean expression is 544 counts in tumor_core_luminal vs 17 in stroma (32-fold enrichment). PGR (510) and GATA3 (665) show the same pattern. Immune infiltration (CD8A 203, CD68 279) is concentrated in the stroma_immune region and largely excluded from the tumor core — an immune-excluded microenvironment. ESR1 Moran's I = 0.42 (significantly clustered, $z=264.7$, $p<0.001$). MKI67 Moran's I = 0.25 (weakly patterned, $z=160.3$) — heterogeneous proliferation across the tumor.

Immune Status: • Low immune activity - The tumor may be evading your immune system

The mapping of gene activity across your tumor tissue shows that estrogen-driven genes dominate the center of your tumor. Immune cells are mostly gathered around the edges rather than inside — your tumor has partially evaded immune surveillance. This confirms hormone-blocking therapy is hitting the right target, and the immune pattern may be relevant if immunotherapy is considered in the future.

Key Patterns Observed

- ESR1 tumor_core_luminal mean 544 vs stroma 17 — 32-fold enrichment, confirming luminal ER-driven biology
- PGR highest in tumor_core_luminal (511) and tumor_proliferative (273) — strong PR+ signal
- GATA3 peak at tumor_core_luminal (665) — supports luminal A/B identity

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- MKI67 spatially heterogeneous (Moran's I 0.25) — proliferation not uniform across tumor

Treatment Options

Based on your test results, your care team may consider these treatments:

Tamoxifen + CDK4/6 Inhibitor (palbociclib, ribociclib, or abemaciclib)

Endocrine therapy
+ targeted

FDA Approved

Expected Response: 60-70% PFS improvement vs. endocrine therapy alone

Your current tamoxifen blocks estrogen from feeding the cancer. Adding a CDK4/6 inhibitor (taken as pills in cycles) further slows cell division. CCND1 amplification and CDKN2A deletion in your tumor both predict strong benefit from this combination.

Why this may help you: ER+/PR+ biology plus CCND1 amplification and CDKN2A deletion — textbook CDK4/6 inhibitor indication

Common side effects: Neutropenia (low white blood cells — requires monthly monitoring), Fatigue, Nausea, Hot flashes from tamoxifen

Olaparib (PARP Inhibitor)

Targeted — PARP inhibitor

FDA Approved

Expected Response: 57-60% ORR in BRCA-mutated HER2- breast cancer (OlympiAD)

Olaparib (pills twice daily) exploits your tumor's BRCA2 DNA repair defect by blocking an alternative repair route. Cancer cells accumulate fatal DNA damage while normal cells cope fine.

Why this may help you: Confirmed pathogenic BRCA2 germline frameshift — FDA-approved indication in germline BRCA1/2-mutated HER2- breast cancer

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Alpelisib + Fulvestrant

Targeted — PI3Ka inhibitor + endocrine

FDA Approved

Expected Response: 35-40% PFS benefit vs. fulvestrant alone (SOLAR-1)

Alpelisib targets the PIK3CA H1047R mutation directly, blocking the overactive growth signal. Combined with fulvestrant (another hormone blocker), it attacks the cancer at two levels simultaneously.

Why this may help you: PIK3CA H1047R at 42% allele frequency — the exact mutation and evidence base studied in the SOLAR-1 FDA approval

Common side effects: Hyperglycemia — blood sugar must be monitored closely, Rash, Diarrhea, Nausea

Clinical Trials

You may be eligible for these research studies offering new treatments:

OlympiA: Adjuvant Olaparib in Germline BRCA-Mutated HER2-Negative Breast Cancer

Phase: 3 | **Status:** Active

Trial ID: NCT03796040

Uses olaparib after surgery for patients with your exact profile — germline BRCA2 mutation and HER2-negative breast cancer. Ask your oncologist if the timing and staging make you eligible.

Contact: clinicaltrials.gov/ct2/show/NCT03796040

INAVO120: Inavolisib + Palbociclib + Fulvestrant in PIK3CA-Mutated HR+ /HER2- Breast Cancer

Phase: 3 | **Status:** Active

Trial ID: NCT05242367

Tests a next-generation PI3K inhibitor targeting your PIK3CA H1047R mutation combined with CDK4/6 inhibitor and fulvestrant — directly relevant to your mutational profile.

Contact: clinicaltrials.gov/ct2/show/NCT05242367

Your Monitoring Plan

Regular check-ups help us track your health and catch any changes early.

Scheduled Tests

Every 6 months	Breast MRI or mammogram Monitor primary tumor response and detect new lesions early
Every 3 months	CEA and CA 15-3 tumor markers Current levels normal (CEA 2.1, CA 15-3 18); rising values signal possible progression
Monthly while on CDK4/6 or PARP inhibitor	Complete blood count (CBC) Detect bone marrow suppression from treatment
Monthly if starting alpelisib	Blood glucose (HbA1c) Alpelisib commonly causes high blood sugar requiring active management
Annually	Pelvic ultrasound and gynecologic exam BRCA2 germline increases ovarian cancer risk — annual surveillance recommended
Within 30 days	Genetic counseling referral BRCA2 germline findings have 50% transmission risk to first-degree relatives

When to Call Your Doctor

Contact your care team right away if you experience:

- New or worsening bone pain — could signal bone metastasis
- Shortness of breath or persistent chest pain
- Severe or escalating headaches

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- Fever above 38°C/100.4°F while on CDK4/6 inhibitor — infection risk with low white cells
- Extreme fatigue or pallor — sign of anemia

Questions? Your oncology nurse navigator or the 24-hour oncology triage line

For Your Family

The BRCA2 mutation is inherited — it is present in every cell of your body and can be passed to your children (50% chance per child). Your parents and siblings each have a 50% chance of carrying it. BRCA2 carriers face a 45-85% lifetime breast cancer risk and 10-20% ovarian cancer risk.

Recommended Steps for Family Members

- Schedule genetic counseling within 4 weeks
- Inform first-degree relatives (parents, siblings, children) and recommend they speak with their doctor about testing
- Discuss risk-reducing surgery options (salpingo-oophorectomy) with your gynecologist
- Consider referral to a hereditary cancer program for long-term surveillance

Genetic Counseling Recommended: A genetic counselor can help explain what these results mean for your family members and discuss testing options.

Resources & Support

You don't have to face this alone. These resources can help:



BRCA Foundation

Education, support, and genetic testing resources for individuals and families with BRCA1/2 mutations

📞 1-800-292-2776 | 🌐 <https://brcafoundation.org>



Susan G. Komen Helpline

Breast cancer helpline, financial assistance programs, and peer community support

📞 1-877-465-6636 | 🌐 <https://www.komen.org>



FORCE — Facing Our Risk of Cancer Empowered

Dedicated support for hereditary breast and ovarian cancer; BRCA-specific peer navigation and counseling referrals

| 🌐 <https://www.facingourrisk.org>

⚠️ Important Notice

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Data Sources: mockepic (DRY_RUN), fgbio (DRY_RUN), genomic-results (fixture data), spatialtools (Visium fixture data), multiomics (Stouffer meta, DRY_RUN p-values)

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